Forecasting U.S. Inflation

Stage 2: October 19 Contribution

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Stage 1 recap

For Stage 2 I use the team's instructor-provided CSVs and Dawson's data loader. They match my September 15 Stage 1 download through 2019. The series are CPIAUCSL, UNRATE, and TB3MS, selecting March, June, September, and December rather than averaging months. Inflation is $\pi_t = 400 \log(P_t/P_{t-1})$, using December 1959 as the base for 1960Q1. There are 160 complete quarters in 1960Q1-1999Q4 and 240 through 2019Q4, with no missing or duplicate quarters.

Inflation averages 4.37%, but is much higher in the 1970s and early 1980s than at the start or end of the sample. The Stage 1 trailing summaries and PELT analysis show a changing center and variability. My unscaled PELT/BIC fit found 17 breaks; this differs from Dawson's six-break, noise-scaled result. The team still needs to agree which version to describe in its combined Step 1. Exact dates are descriptive, not confirmed regime boundaries, and do not enter the forecasting regressions below. The inflation-unemployment correlation is 0.10, so that scatter alone does not tell us whether unemployment improves forecasts.

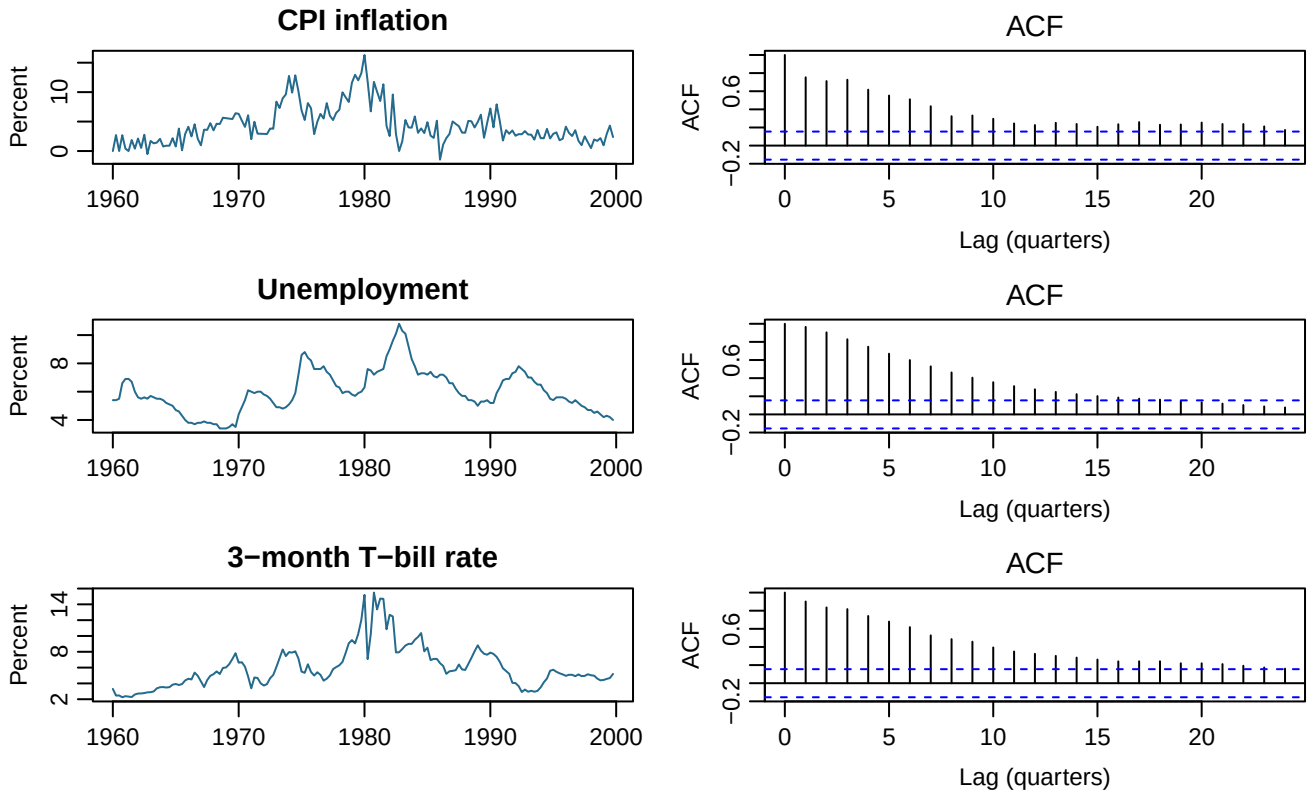

Figure 1: Initial-sample time plots and ACFs. Lag is measured in quarters; dashed ACF bands are approximate 95% white-noise bounds.

Step 1: Stationarity and structural change

All three series are persistent in Figure 1. Their lag-1 autocorrelations are 0.75, 0.97, 0.90, respectively. Inflation and the bill rate show particularly large changes around the high-inflation period; unemployment moves through long cycles, including its early-1980s peak. None looks well described by one straight deterministic trend.

I therefore use an intercept but no linear trend in each ADF regression, and level-stationarity KPSS tests. The ADF null is a unit root; the KPSS null is stationarity around a constant. I compare both tests at 5%. For ADF, I select 0-4 augmentation lags by BIC on a common sample, then refit the selected lag using all available rows. For KPSS, I use the short bandwidth, which is four lags here. The same choices apply to first differences.

Series	Form	ADF lags	ADF	KPSS
CPI inflation	Level	2	-2.272	0.500
CPI inflation	First difference	1	-15.237	0.075
Unemployment	Level	1	-2.379	0.541
Unemployment	First difference	0	-7.409	0.098
3-month T-bill rate	Level	3	-2.440	0.646
3-month T-bill rate	First difference	2	-7.233	0.075

The 5% critical values are -2.880 for ADF and 0.463 for KPSS. Reject the ADF null for a more negative statistic, and the KPSS null for a larger statistic. At levels, ADF does not reject a unit root for any series, while KPSS rejects constant-mean stationarity for all three. After one difference, ADF rejects and KPSS does not reject for every series.

Series	ADF evidence	KPSS evidence	Structural-change evidence	Classification
CPI inflation	Reject only after differencing	Reject only in levels	Changing mean and variability	$I(1)$, provisionally
Unemployment	Reject only after differencing	Reject only in levels	Long cycles; early-1980s peak	$I(1)$, provisionally
T-bill rate	Reject only after differencing	Reject only in levels	Early-1980s peak, then decline	$I(1)$, provisionally

These are working classifications, not proof of a permanent unit root. A changing mean can make stationary behavior within periods look nonstationary over the full sample, and the tests do not separate those explanations. The project still calls for its specified forecasting models; I would not replace them with differenced models solely because of these tests. The changing inflation mean gives a reason to compare the gap models with fixed-mean alternatives.

Step 2: Expanding-window forecast setup

The first forecast origin is 1999Q4, using 1960Q1-1999Q4 for estimation. At each later origin, I add one quarter, re-estimate the model, and retain forecasts at horizons 1, 4, and 8. The final historical target is 2019Q4; observations from 2020 onward are reserved for the later project stage.

Horizon	First origin	Last origin	First target	Last target	Errors
1	1999Q4	2019Q3	2000Q1	2019Q4	80
4	1999Q4	2018Q4	2000Q4	2019Q4	77
8	1999Q4	2017Q4	2001Q4	2019Q4	73

The schedule contains 230 origin-horizon pairs per model. The counts differ because a forecast made eight quarters ahead reaches its first target later than a one-quarter forecast. Within each horizon, all seven models must use the same target quarters.

Estimation and data boundaries

At origin T, only observations through T enter fitting, lag selection, or smoothing. For my direct AR and direct Phillips-curve models, the response is π_{t+h}, so a training row is usable only when $t + h \leq T$. Each horizon gets its own regression and BIC lag selection over 1-4 inflation lags. Candidate lag orders should be compared on the same available response rows.

Variables, transformations, candidate lag ranges, estimation methods, horizons, and forecast construction stay fixed throughout evaluation; only fitted parameters and BIC-selected lag orders are updated at each origin. Later forecast errors must not be used to revise those choices.

For example, at origin 1999Q4 with $h = 4$, the latest training row is dated 1998Q4 and its response is inflation in 1999Q4. The new forecast uses regressors observed in 1999Q4 to predict inflation in 2000Q4. The Phillips-curve model adds unemployment observed at the regressor date, not future unemployment.

The source files contain revised historical observations rather than real-time vintages, so the experiment is pseudo-out-of-sample, not a reconstruction of the information actually released at each historical date.

Shared output for the model comparison

My model functions return a three-row tibble with `h` and `forecast`, as required by the shared harness. The harness adds `model`, `origin`, `target`, `actual`, and `error = actual - forecast`. Positive errors mean underprediction. Both models pass its count and no-lookahead checks. The saved forecasts contain 230 rows per model, with no missing or duplicate origin-horizon pairs.

Step 3: Models 4 and 7

Model 4: Direct AR

For each horizon $h = 1, 4, 8$, I fit a separate regression with an intercept:

$$\pi_{t+h} = c_h + \sum_{j=0}^{p_h - 1} \phi_{h,j} \pi_{t-j} + \varepsilon_{t+h}.$$

At origin T, the final response in estimation is π_T, so the most recent regressor row is dated $T - h$. BIC selects p_h from 1 through 4 on a common sample beginning at the fourth training quarter. I then refit the selected order on all rows available for that order. The forecast uses inflation at T and its selected lags. No iterative forecasts or future inflation regressors are used.

Model 7: Direct Phillips curve

The Phillips-curve model adds unemployment observed at each regressor date:

$$\pi_{t+h} = c_h + \sum_{j=0}^{p_h - 1} \phi_{h,j} \pi_{t-j} + \lambda_h u_t + \varepsilon_{t+h}.$$

It has its own BIC lag selection at each horizon and origin, using the same candidate orders and sample rule as the direct AR. Forecasting uses u_T, not unemployment at $T + h$. Both models stay in levels, as the guide specifies.

The code in `R/03_models_odysseus.R` supplies `direct_ar_model(train)` and `phillips_model(train)`. The small shared fitting function keeps their response alignment and lag-selection rules identical.

Table 1: Forecasts from the first origin, 1999Q4 (annualized percent).

Model	Horizon	Target	Lags	Forecast
Direct AR	1	2000 Q1	3	3.112
Phillips curve	1	2000 Q1	3	3.649
Direct AR	4	2000 Q4	1	3.176
Phillips curve	4	2000 Q4	2	3.956
Direct AR	8	2001 Q4	1	3.842
Phillips curve	8	2001 Q4	1	3.818

The target is inflation in the indicated single quarter, not average or cumulative inflation over the next h quarters. Coefficients and selected lag orders are updated as the estimation window expands; the model definitions do not change.

Sources

- W271, *Forecasting U.S. Inflation: Project Guide*, Stage 2, Steps 1-3; my Stage 1 report; and the team's shared data loader and forecast harness.
- Hyndman and Athanasopoulos, *Forecasting: Principles and Practice*, stationarity and time-series cross-validation.
- R `urca` documentation for ADF and KPSS.
- R documentation for BIC.